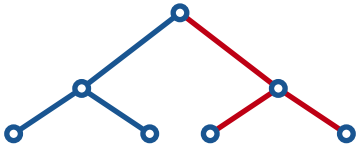
Chapter 14

Details of the Extensive Form



Defining an Extensive-Form Game

An **extensive-form game** specifies the players, how play unfolds, what each player knows when acting, and the payoffs from every possible outcome. Here we consider games with finitely many stages, while allowing continuous action sets.

- The game identifies a set of **players** and a rooted **game tree**. Play begins at the initial node. Each branch extends the history by one move, and play ends at a terminal node.
- Each nonterminal node specifies who moves: a player or Nature. Its outgoing branches represent the available actions or chance outcomes, each of which leads to a distinct successor node.
- At a **chance node**, Nature selects an outcome according to a probability distribution specified by the game. These probabilities are fixed; Nature does not choose them strategically.

Information and Payoffs Complete the Game

An extensive-form game also specifies what each player knows when acting and how each player evaluates outcomes. We describe these features through information sets and payoffs.

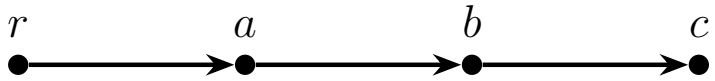
- For each player, **information sets** partition that player's decision nodes. Nodes in the same set are indistinguishable to the player when acting and must offer the same labeled actions. A singleton set means the player knows the exact history.
- Each terminal node assigns a **payoff** to every player. When play involves random outcomes, players evaluate the resulting lotteries using expected payoffs. Nature has no payoff.

Once the game is specified, a player can formulate a complete plan of action. Such a plan is a pure strategy: it prescribes an action at every information set of that player, including those that play may never reach.

I. Nodes, Branches, and Precedence

A game tree records the sequence of player decisions and chance events. A node reached by following branches from another node is its **successor**; a node reached in one branch is an **immediate successor**.

The diagram illustrates these relations. Node b is an immediate successor of a, while c is a successor of a reached through b.



Reading backward, a is a **predecessor** of c, and b is its **immediate predecessor**. The initial node has no predecessor; a terminal node has no outgoing branches.

A Tree Gives Each History a Unique Location

Two rules ensure that every node has a unique history from the start.

Tree Rule 1. Every node other than the initial node can be reached from that initial node by following branches.

Tree Rule 2. Every node except the initial node has exactly one immediate predecessor. The initial node has none.

Together, these rules prevent a tree from merging two different histories into one node or returning to an earlier node in a cycle.

Two different terminal nodes may have identical payoffs. Their positions still distinguish the histories that led to those payoffs.

Paths, Actions, and Terminal Payoffs

A **path** starts at the initial node, follows branches through successive immediate successors, and ends at a terminal node. Every terminal node identifies a unique path through the tree.

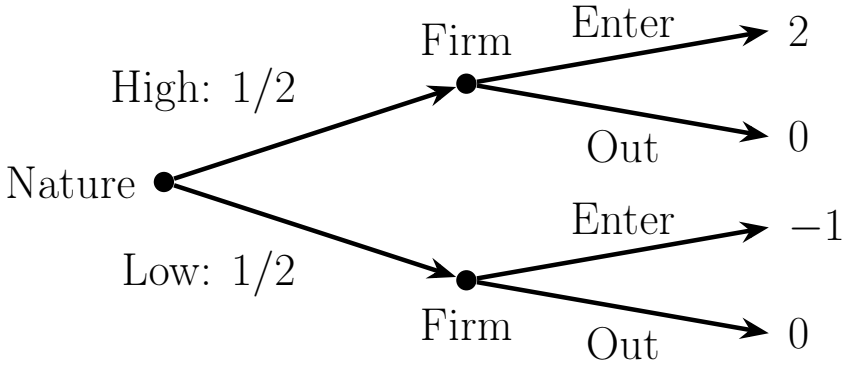
Tree Rule 3. Branches leaving the same nonterminal node must have different action or outcome labels. Each label identifies a unique next node.

Label each nonterminal node with the player who moves or with Nature if chance determines the next branch. For finitely many chance outcomes, specify nonnegative branch probabilities that sum to one.

Each terminal node lists payoffs in player order: player 1 first, then player 2, and so on. Nature has no payoff.

Chance Moves Are Moves by Nature

Nature represents uncertainty outside the players' control. For example, demand is high or low with equal probability; a firm observes demand before choosing whether to enter. Terminal numbers are the firm's payoffs.

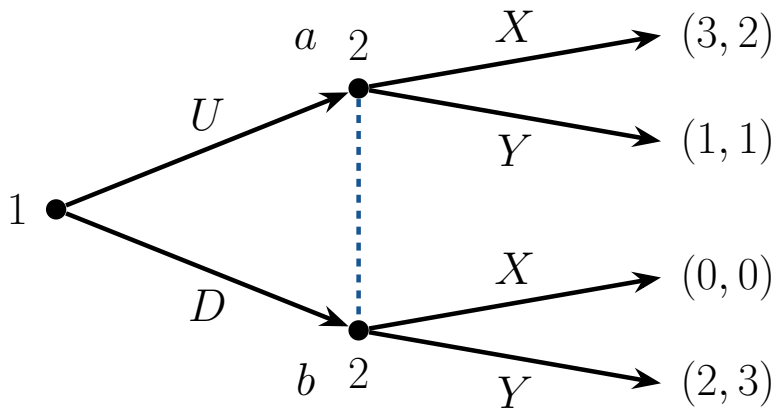


A pure strategy specifies the firm's action after each demand outcome. Nature's probabilities remain fixed. Always entering gives expected payoff $\frac{1}{2}(2) + \frac{1}{2}(-1) = \frac{1}{2}$.

Entering only after high demand gives expected payoff $\frac{1}{2}(2)+\frac{1}{2}(0) = 1$. Even with a fixed pure strategy, the realized path depends on Nature's move.

II. Information Sets Describe What Is Known

Player 1 chooses U or D. Player 2 then chooses X or Y without observing that choice. Nodes a and b therefore share an information set.



The dashed line means that player 2 cannot distinguish a from b. A pure strategy must prescribe the same action at both nodes.

One Player and One Menu

An information set must describe one player facing one action menu.

Tree Rule 4. All nodes in an information set belong to the same player. The set describes that player's uncertainty when choosing.

Tree Rule 5. Every node in an information set has the same available actions, with the same labels and number of outgoing branches.

- For example, a player cannot face only X, Y at one node and X, Y, Z at another node in the same information set. The available menu would reveal which node had been reached, and a plan choosing Z could not be implemented at both nodes.

Perfect Information and Perfect Recall

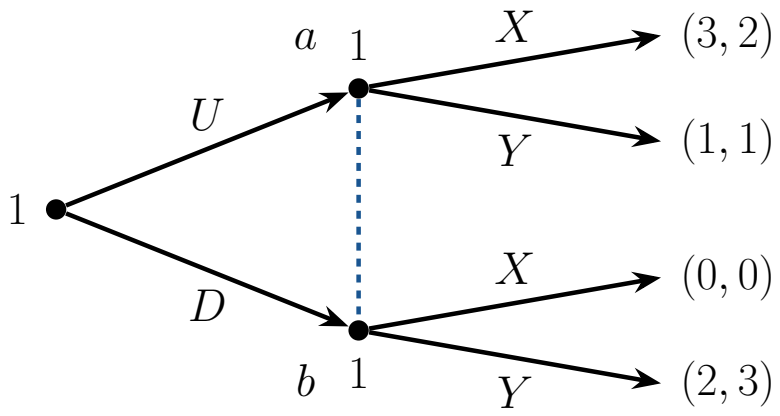
A game has **perfect information** if every player information set contains one decision node. At each decision, the player knows the entire history, including past chance outcomes. Otherwise, the game has **imperfect information**.

A game has **perfect recall** if each player remembers their own past actions and the information they previously had. Perfect recall allows uncertainty about actions or chance outcomes never observed.

- In the preceding U, D game, player 2 has imperfect information because player 1's action is hidden. Both players have perfect recall: neither has forgotten a past observation or an action of their own.

Forgetting an Earlier Choice

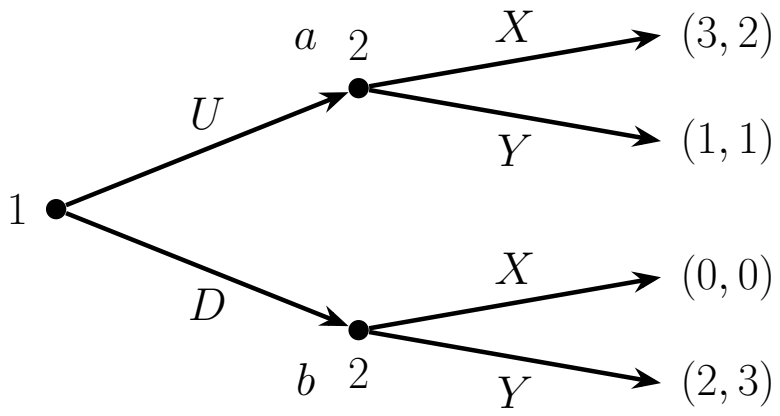
Now let player 1 make both decisions; player 2 receives a payoff but never moves. Joining a and b means that player 1 has forgotten whether they chose U or D.



The action menus match, so the information set satisfies Rules 4 and 5. However, the game violates perfect recall. Perfect recall is an additional restriction on what players remember.

Observation Changes the Strategy Space

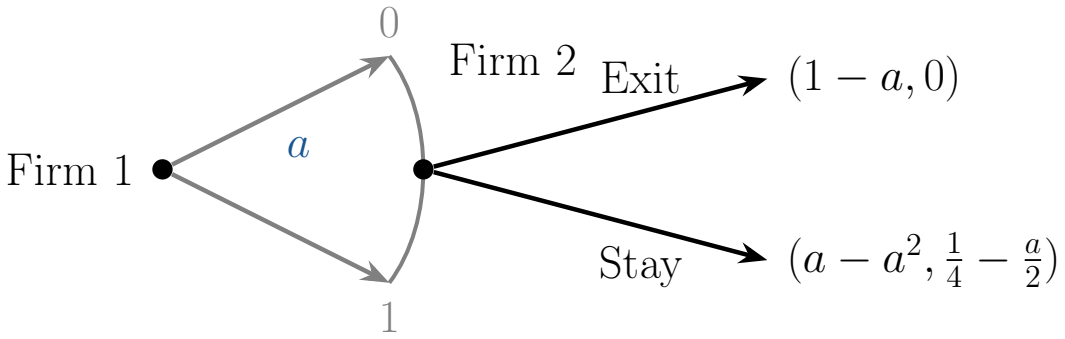
Return to the game where player 2 moves after player 1. If player 2 observes U or D, nodes a and b are separate information sets. Player 2 can then condition their choice on player 1's action.



Player 2 now has four pure strategies: XX, XY, YX, YY, listing the action after U first and after D second. Without observation, player 2 has only two pure strategies, X and Y.

III. Drawing a Continuum of Actions

Firm 1 chooses advertising $a \in [0, 1]$. Firm 2 observes a and decides whether to Exit or Stay. The payoffs below are given functions of a.



The arc represents every advertising level between 0 and 1. The displayed continuation is a generic one: there is a separate decision node for firm 2 after each observed value of a.

A Strategy for Observed Advertising

With observed advertising, firm 2's pure strategy specifies a decision for every possible advertising level:

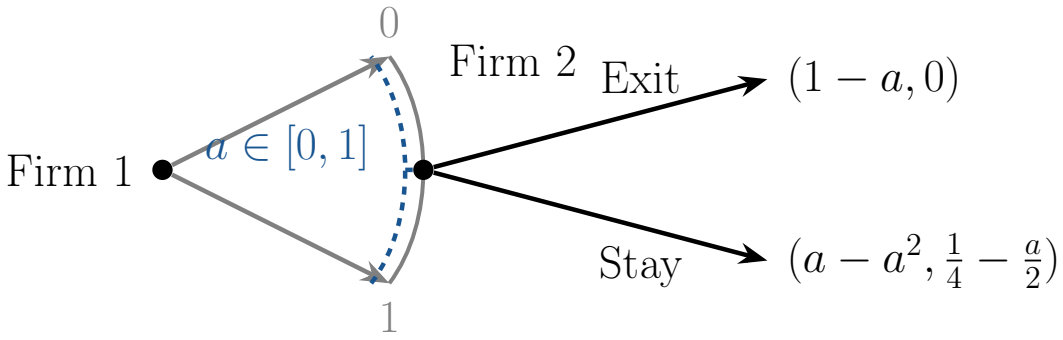
$$s_2 : [0, 1] \longrightarrow \{\text{Exit}, \text{Stay}\}.$$

- For example, a cutoff plan can prescribe Stay when $a \leq 1/2$ and Exit when $a > 1/2$, with Stay prescribed at the cutoff itself.

Each value of a leads to a different singleton information set. Firm 2 can therefore implement this plan by conditioning its action on the observed advertising level.

A Continuum of Hidden Advertising Choices

Now suppose firm 2 cannot observe advertising. All of its decision nodes belong to one information set, shown by the dashed curve beside the arc.

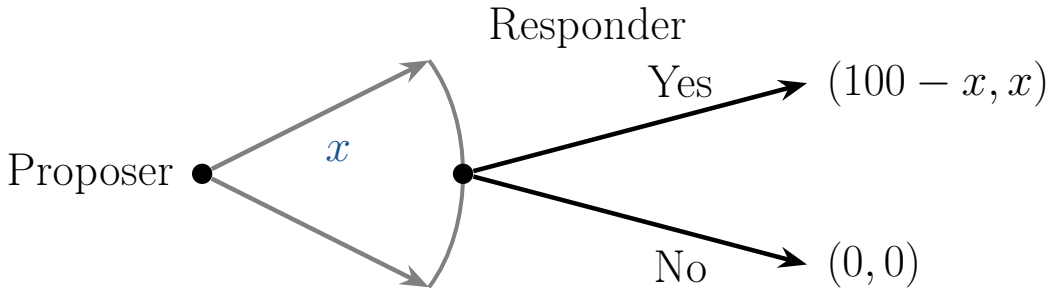


The displayed continuation represents the same Exit/Stay menu at every value of a. Firm 2 must choose the same action throughout the information set, so its pure strategy is simply Exit or Stay.

Changing observation changes the strategy space even when the physical actions and payoff formulas remain the same.

IV. Ultimatum-Offer Bargaining

A proposer and a responder can divide 100 monetary units. The proposer offers $x \in [0, 100]$ to the responder, who observes x and either accepts or rejects. Acceptance gives $100 - x$ to the proposer and x to the responder; rejection gives both players zero.



The payoff pair lists the proposer's payoff first and the responder's payoff second. At $0 < x < 100$, both players receive more from acceptance than from rejection.

An Action at One Node Is Part of a Strategy

The proposer moves once, so a pure strategy is a single offer:

$$S_1 = [0, 100].$$

The responder observes the offer and has one information set for each possible offer. A pure strategy must specify a response at every offer:

$$s_2 : [0, 100] \longrightarrow \{\text{Yes}, \text{No}\}.$$

- Accepting an observed offer of 50 describes the responder's action at that information set. A complete strategy also specifies what the responder would do after offers of 20, 80, or any other feasible offer.

Examples of Complete Responses

The responder could accept every offer or reject every offer. Another strategy is a cutoff rule: for a chosen $\bar{x} \in [0, 100]$,

$$s_2(x) = \begin{cases} \text{Yes}, & x \geq \bar{x}, \\ \text{No}, & x < \bar{x}. \end{cases}$$

- With $\bar{x} = 50$, this strategy accepts 50 and 80 but rejects 20. The equality specifies acceptance at the cutoff itself.

Other functions are also valid strategies. For example, accepting only offers of 20 and 80 is a complete plan once rejection at every other offer is specified. Defining a strategy does not require it to be optimal.

Responses at Unreached Information Sets

Responses to unmade offers matter because the proposer can deviate. Suppose the proposer plans to offer 50. A responder who accepts every offer and a responder who accepts only offers of at least 50 both accept that offer. Their behavior differs if the proposer changes the offer.

To evaluate a deviation to 20, the proposer must know whether the responder's plan accepts or rejects 20. The response at this initially unreached information set affects the proposer's incentives at the start.

Complete plans also prescribe behavior when someone makes a mistake or departs from an earlier plan. A pure strategy therefore specifies an action at every information set belonging to the player.

V. From Strategies to Paths and Payoffs

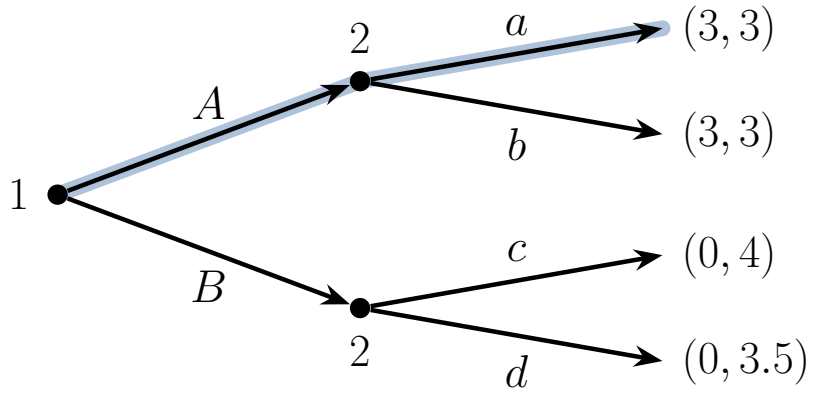
With no chance moves, a pure-strategy profile determines the sequence of actions. At the initial node, the acting player's strategy chooses one branch. At the next decision node, the relevant strategy again chooses one branch.

Continuing this procedure determines a unique path to a terminal node in a game with finitely many stages. The terminal payoff vector gives each player's payoff from the full strategy profile.

Different profiles can induce the same path because their prescriptions may differ only at information sets that the path never reaches. With chance moves, combine the profile with Nature's given probabilities to obtain a distribution over terminal paths. Players evaluate the profile using expected payoffs over that distribution.

Different Plans Can Produce the Same Path

In this game, player 2 observes player 1's action. A strategy for player 2 lists an action after A and an action after B.



Both (A, ac) and (A, ad) induce the highlighted path (A, a). The two profiles differ in player 2's response after B.

The Normal Form Records Complete Plans

For this game without chance moves, each pair of complete plans induces a path and its terminal payoffs. The corresponding normal form is:

Player 1 / Player 2	ac	ad	bc	bd
A	$3, 3$	$3, 3$	$3, 3$	$3, 3$
B	$0, 4$	$0, 3.5$	$0, 4$	$0, 3.5$

Here $S_1 = \{A, B\}$ and $S_2 = \{ac, ad, bc, bd\}$. Player 2 has two actions at each of two information sets, giving $2 \cdot 2 = 4$ pure strategies. The columns distinguish plans even when they produce the same path.

With chance moves, each cell instead records expected payoffs over the paths induced by the strategies and Nature's probabilities.

Nash Equilibrium in Complete Plans

A pure-strategy profile in an extensive-form game is a Nash equilibrium if no player can gain by changing their complete plan:

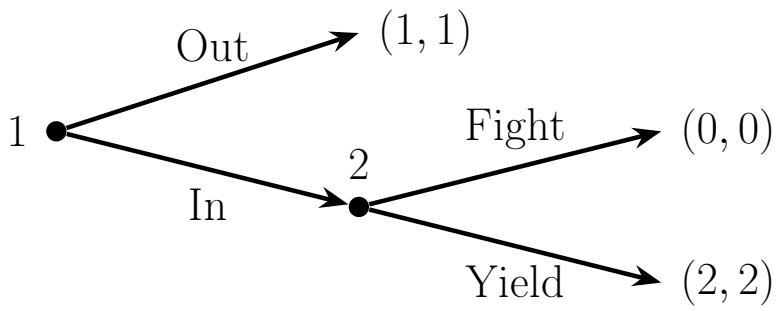
$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \qquad \text{for every player } i \text{ and feasible } s_i.$$

Here u_i is player i's payoff, evaluated in expectation when Nature moves. Other players' plans and Nature's probabilities remain fixed. A deviation can change which information sets are reached, so responses at previously unreached information sets can affect its payoff.

Consequently, observing the path alone may be insufficient to determine whether the underlying strategies form a Nash equilibrium.

Unreached Responses Affect Incentives

Player 1 chooses Out or In. After In, player 2 chooses Fight or Yield. The terminal payoffs are ordered as (player 1, player 2).



Both (Out, Fight) and (Out, Yield) end at Out with payoff $(1, 1)$. The two plans differ in player 2's response if player 1 enters.

Checking the Two Profiles

Let's check whether the two plans that lead to Out are equilibria. At (Out, Fight), player 1 receives 1. Switching to In would give 0, so player 1 prefers Out. Player 2's choice is never reached, so changing Fight to Yield leaves player 2's payoff at 1. This profile is a Nash equilibrium.

At (Out, Yield), player 1 can switch to In and raise their payoff from 1 to 2. This profile is not a Nash equilibrium, even though it generates the same observed path.

The plan to Fight supports the first equilibrium. If the decision after In were reached, player 2 would prefer Yield because $2 > 0$. Nash equilibrium permits this prescription at an unreached information set.

VI. An Equilibrium Offer of 50

Return to the division of 100 monetary units. Suppose the proposer offers $x = 50$ and the responder uses the cutoff rule $\bar{x} = 50$. The offer is accepted, giving payoffs $(50, 50)$.

Holding the cutoff rule fixed, a higher offer leaves the proposer less than 50. A lower offer is rejected and gives zero. So the proposer cannot profit by changing the offer.

Holding the offer fixed at 50, acceptance gives the responder 50 and rejection gives zero. Changing responses at other offers has no effect on this payoff. The two strategies therefore form a Nash equilibrium.

Supporting Any Offer in Equilibrium

The same cutoff construction supports any offer $x^* \in [0, 100]$. Let the proposer offer x^* and the responder accept exactly when $x \geq x^*$. Acceptance gives payoffs $(100 - x^*, x^*)$.

A proposer who raises the offer keeps less, while a lower offer is rejected and gives zero. Since $100 - x^* \geq 0$, no deviation improves the proposer's payoff. The responder receives $x^* \geq 0$ from acceptance, so rejecting the offer also cannot improve their payoff.

- The endpoints are included: at $x^* = 100$ the proposer is indifferent to lower rejected offers, and at $x^* = 0$ the responder is indifferent between acceptance and rejection of the offer.

Unreached Responses Support Different Offers

In the equilibrium with offer 50, the responder's strategy rejects an offer of 20. This response deters the proposer from reducing the offer.

If 20 were actually offered, accepting would give the responder 20 while rejecting would give zero. Nevertheless, changing this response alone has no effect when the proposer's fixed strategy is to offer 50.

The Nash condition evaluates a deviation against the other player's whole strategy. It can therefore leave a player indifferent between plans that differ only at unreached information sets, even when one of those prescriptions would be undesirable if that set were reached.

VII. Quantity Choices after Observing a Rival

Firm 1 chooses $q_1 \in [0, 12]$. Firm 2 observes that choice and then chooses $q_2 \in [0, 12]$. Market price and profits are

$$P = 12 - q_1 - q_2, \qquad \pi_i = (12 - q_1 - q_2)q_i.$$

Production costs are zero. The linear price formula applies to every feasible pair of quantities, including pairs that give a negative price.

Firm 1's pure strategy is a number. Firm 2's pure strategy is a function

$$s_2 : [0, 12] \longrightarrow [0, 12], \qquad q_2 = s_2(q_1).$$

Specifying only a realized quantity for firm 2 leaves its responses to other possible quantities of firm 1 undefined.

A Complete Plan That Discourages Deviation

A response that drives price to zero after a deviation can sustain $q_1 = 2$ in Nash equilibrium. Consider this strategy for firm 2:

$$s_2(q_1) = \begin{cases} 5, & q_1 = 2, \\ 12 - q_1, & q_1 \neq 2. \end{cases}$$

At the induced quantities $(2, 5)$, price is 5 and profits are $(10, 25)$. If firm 1 changes its quantity, firm 2's prescribed response makes price zero, so firm 1 earns zero. Firm 1 therefore prefers to keep $q_1 = 2$.

Against the fixed quantity $q_1 = 2$, firm 2's profit is

$$(10 - q_2)q_2 = 25 - (q_2 - 5)^2,$$

which is maximized at $q_2 = 5$. Changing its responses at other quantities has no effect. Thus, the profile is a Nash equilibrium.

A Family of Equilibria in Quantities

The same construction supports any leader quantity $x \in [0, 12]$. Let firm 1 choose $q_1 = x$ and firm 2 use

$$s_2(q_1) = \begin{cases} (12 - x)/2, & q_1 = x, \\ 12 - q_1, & q_1 \neq x. \end{cases}$$

At x, firm 2's prescribed quantity maximizes $(12-x-q_2)q_2$. Firm 1 receives $x(12-x)/2 \geq 0$; every deviation gives zero because the prescribed response drives price to zero. These strategies form a Nash equilibrium, including at $x = 0$ and $x = 12$.

In any pure-strategy Nash equilibrium, firm 2 must choose $(12-q_1)/2$ at the realized q_1: changing that response alone would otherwise raise its profit. Its unreached responses may differ.

Takeaways

An extensive-form game specifies how play unfolds and what players know.

- The tree records who moves, available actions, chance probabilities, and terminal payoffs. Information sets describe what players can distinguish.
- Perfect information means knowing the full history; perfect recall means remembering one's own past actions and information.

Nash equilibrium compares complete plans, including unreached responses.

- A pure strategy prescribes an action at every information set. A profile determines a path or, with chance moves, a distribution over paths and expected payoffs.
- Unreached responses can deter deviations even when they would be undesirable if the relevant information set were reached.